Impact of Baseline Hard Drug Use on HAART Treatment Response

A Frequentist and Bayesian Analysis of the MACS Cohort

Mahfuza Haque Mahi

Biostatistics PhD Student, Colorado School of Public Health
University of Colorado Anschutz Medical Campus

BIOS 6624 – Project 1

Introduction & Research Objectives

Background

HAART substantially improves outcomes for HIV-positive individuals. However, baseline hard drug use may impair treatment response through direct immunotoxic effects and reduced treatment adherence.

Primary Objective: Is baseline hard drug use associated with Year 2 outcomes in HIV-positive men (MACS cohort)? **Four outcomes at Year 2:**

- Viral load ($\log_{10}$ copies/mL)
- $CD4^+$ T-cell count (cells/μL)
- Physical QOL (SF-36, 0–100)
- Mental QOL (SF-36, 0–100)

Secondary Objective: Does adjusting for medication adherence ($\geq$95% vs. $<$95%) change the estimates?

Methods

Study Population

- HIV-positive men, MACS cohort; **N = 463**
- Hard drug users: $n = 36$ (7.8%)
- everART $= 1$; complete Year 0 & Year 2 data
- Covariates: age, BMI, race, education, smoking

ANCOVA Model (8 models: 4 outcomes $\times$ 2)

$$Y_2 = \beta_0 + \beta_1 \text{DrugUse} + \beta_2 Y_0 + \mathbf{X}\boldsymbol{\beta} + \varepsilon$$

- Unadjusted & adherence-adjusted versions

Bayesian Approach (Stan / HMC-NUTS):

- Priors: $\beta \sim \mathcal{N}(0, 100^2)$, $\sigma \sim$ Half-$\mathcal{N}(0, 100)$
- 4 chains $\times$ 2,000 iterations; R-hat < 1.01

Bayesian Summaries:

- Posterior mean & 95% HPDI
- P($\beta < 0$): probability of harmful effect
- **P(Clin.)**: probability effect exceeds clinical threshold
  - Viral load: 0.5 $\log_{10}$ copies/mL
  - CD4: 50 cells/μL; QOL: 2 SF-36 pts
- LOO-IC for model comparison

Descriptive Statistics & Outcome Trajectories

Table 1: Baseline Characteristics (n = 463)

Variable	No Drugs (n = 427)	Hard Drugs (n = 36)
Age (years), mean (SD)	43.1 (8.68)	43.9 (9.52)
BMI (kg/m^2), mean (SD)	25.3 (4.39)	23.6 (3.45)
Race		
White Non-Hispanic	277 (64.9%)	19 (52.8%)
Other	150 (35.1%)	17 (47.2%)
Education		
HS or less	84 (19.7%)	13 (36.1%)
Some college	152 (35.6%)	13 (36.1%)
College+	191 (44.7%)	10 (27.8%)
Smoking status		
Current	149 (34.9%)	**27 (75.0%)**
Not current	278 (65.1%)	9 (25.0%)
Viral load, mean (SD)	4.53 (0.93)	4.58 (0.87)
CD4 count, mean (SD)	374 (202)	360 (201)
Physical QOL, mean (SD)	51.4 (9.00)	49.2 (7.05)
Mental QOL, mean (SD)	45.2 (13.8)	42.0 (11.6)
Adh. $\geq$95% (Yr 2)	382 (89.5%)	35 (97.2%)

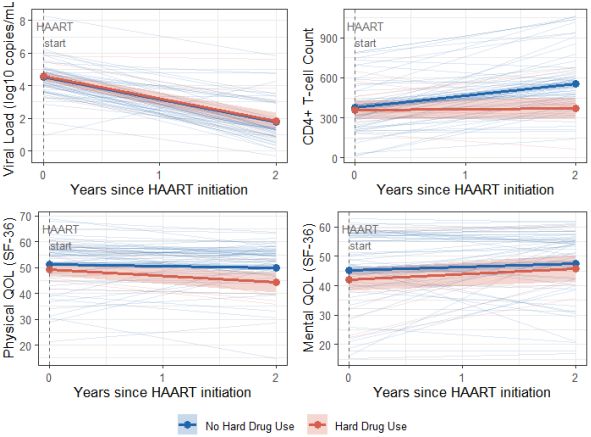

Fig. 1. Mean outcome trajectories ± 95% CI by hard drug use group.

Results: Effect of Baseline Hard Drug Use

Frequentist: Est (95% CI)

	Unadjusted		+ Adherence
Outcome	**Est (95% CI)**	**p**	**Est (95% CI)**
Viral Load	-0.06 $(-0.46, 0.34)$	0.761	-0.01 $(-0.41, 0.38)$
CD4 Count	-163.97 $(-226.54, -101.41)$	**<0.001**	-169.86 $(-232.50, -107.2$
Phys. QOL	-3.25 $(-5.97, -0.53)$	**0.019**	-3.48 $(-6.20, -0.75)$
Mental QOL	-0.28 $(-3.74, 3.19)$	0.876	-0.52 $(-4.00, 2.96)$

Bayesian: Posterior Mean (95% HPDI)

	Unadjusted		+ Adherence
Outcome	**Mean (95% HPDI)**	**P(β¡0)**	**Mean (95% HPDI)**
Viral Load	-0.06 $(-0.44, 0.32)$	0.621	-0.01 $(-0.41, 0.41)$
CD4 Count	-146.49 $(-206.82, -88.18)$	**1.000**	-152.79 $(-214.23, -94.84)$
Phys. QOL	-3.26 $(-5.95, -0.52)$	**0.993**	-3.48 $(-6.22, -0.79)$
Mental QOL	-0.26 $(-3.75, 3.19)$	0.557	-0.53 $(-4.01, 2.99)$

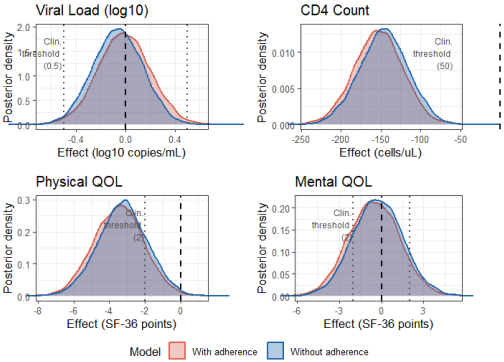

Fig. 2. Posterior densities for the hard drug use coefficient. Dashed = no effect; dotted = clinical threshold.

Results: Adherence Effect & Model Fit

Table 3: Adherence $\geq$95% vs. $<$95%

Outcome	Freq. Est (95% CI)	p	Bayes Mean (95% HPDI)	P(β¡0)	P(Clin.)
Viral Load	-0.44 (-0.79, -0.09)	**0.014**	-0.44 (-0.78, -0.08)	0.993	0.362
CD4 Count	$+55.54$ (0.49, 110.58)	**0.048**	$+54.51$ (1.92, 106.29)	0.021	0.569
Phys. QOL	$+2.13$ (-0.26, 4.52)	0.081	$+2.13$ (-0.10, 4.58)	0.039	0.543
Ment. QOL	$+2.30$ (-0.76, 5.36)	0.139	$+2.32$ (-0.59, 5.39)	0.063	0.586

Table 4: LOO-IC Model Comparison

Outcome	No Adh.	With Adh.	ΔLOO-IC
Viral Load	1441.6	1438.4	-3.2
CD4 Count	6126.2	6123.6	-2.6
Physical QOL	3221.9	3220.2	-1.7
Mental QOL	3446.1	3446.3	$+0.2$

Key Takeaways

- Higher adherence $\Rightarrow$ significantly **lower viral load** & **higher CD4**
- P($\beta < 0$) = 0.993 for viral load
- P(Clin.) $>$ 0.5 for CD4 & all QOL outcomes
- Bayesian & frequentist estimates aligned
- Modest LOO-IC improvement for lab outcomes
- No ΔLOO-IC exceeded 4

Discussion & Conclusions

Key Findings

- Hard drug use $\Rightarrow$ worse immune recovery: CD4: -147 cells/μL, P($\beta < 0$) $= 1.00$, P(Clin.) $= 1.00$
- Physical QOL: -3.26 pts, P($\beta < 0$) $= 0.993$, P(Clin.) $= 0.993$
- Both effects **persist after adherence adjustment**
- Adherence independently improves viral load & CD4; P(Clin.) > 0.5 for both
- Bayesian uniquely quantifies clinical relevance — P(Clin.) $= 1.00$ for CD4 means the entire posterior is past the 50-cell threshold
- LOO-IC: modest improvement with adherence for lab outcomes

Limitations

- Small hard drug user group ($n = 36$)
- Complete-case analysis: potential selection bias
- Drug use measured at baseline only

Clinical Implication

Addressing **substance use** and **adherence** *concurrently* is critical to optimize immune recovery following HAART initiation.

Code: github.com/mhaque-99/BIOS6624